

New York University Tandon School of Engineering

Department of Computer Science & Engineering

Introduction to Operating Systems – CS-UY 3224

Fall 2026

Professor Gustavo Sandoval

Tue/Th – 2:00 pm – 3:50 pm

Contact`

Email: gustavo.sandoval@nyu.edu

My background is [here](#).

Office hours:

- Zoom Room: [nyu.zoom.us/my/Sandoval](https://nyu.zoom.us/j/9111111111)
- Office hours: Wednesday 2 - 3

Communication:

- We will be using EdStem integrated with Brightspace. Should be there shortly.

Student Assistants:

See EdStem for their student hours. They will be posted shortly.

Course Pre-requisites

CS-UY 2214 (Computer Architecture and Organization)

CS-UY 2134 (Data Structures and Algorithms)

Strong Programming in C or C++. (You will have a tough time, and are at risk of failing the class if you haven't programmed in C or C++ before)

Course Description

This course is an introduction to operating system design and implementation. We study operating systems because they are examples of mature and elegant solutions to a complex design problem: safely and efficiently sharing system resources and providing abstractions helpful to applications.

For the processor, memory, and disks, we discuss how the operating system allocates each resource and explore the design and implementation of related abstractions.

We also establish techniques for testing and improving system performance and introduce the idea of hardware virtualization. Programming assignments provide hands-on experience implementing core operating system components in a realistic development environment. We will examine the design and implementation of a UNIX-like operating system in detail and cover general operating systems concepts, such as processes, threads, memory, virtual memory, device drivers, filesystems, scheduling, concurrency, security, and virtualization.

Expected Learning Outcomes

ABET Outcome	OS Course Outcome
1. Analyze a complex computing problem...	Analyze and compare OS algorithms (e.g., scheduling, memory, file systems), identifying trade-offs in performance and resource use.
2. Design, implement, and evaluate a computing-based solution...	Build or modify OS components (e.g., process scheduler, file system) using appropriate tools and techniques (e.g., C, gdb, system calls).
3. Communicate effectively...	Clearly explain OS design decisions and system behavior in written reports or presentations.
4. Recognize professional responsibilities...	Understand ethical and security implications of OS design (e.g., resource isolation, access control, privilege escalation).
5. Function effectively as a team...	Collaborate on OS projects, such as building a kernel module or debugging concurrency issues.
6. Apply computer science theory and software development fundamentals...	Apply concepts like synchronization, virtual memory, and file system organization in real or simulated OS environments.

Readings

Textbooks:

- Arpaci-Dusseau, R and Arpaci-Dusseau A. **Operating Systems Three Easy Pieces**. Available [online](#) for free.
- Russ Cox, Frans Kaashoek, and Robert Morris, **xv6: A simple, Unix-like teaching operating system**. <http://pdos.csail.mit.edu/6.828/2014/xv6/book-rev8.pdf>
- Andrew S. Tannenbaum, **Modern Operating Systems**. Available on NYU Classes.

Useful references:

- Linux command line cheat sheet:

[File:Unix command cheatsheet.pdf - Wikimedia Commons](#)

- Gdb cheat sheet:
<http://darkdust.net/files/GDB%20Cheat%20Sheet.pdf>

Participation (10%)

We will use Poll Everywhere for in-class participation. You earn full credit if you respond to at least 80% of the counted participation polls over the semester. Below 80%, participation credit scales proportionally. There are **no makeups for missed polls**; the **20% cushion covers absences and technical issues**. To ensure credit, access Poll Everywhere through Brightspace at least once during the first week.

Cooperation Policy

You will work individually on every assignment, unless otherwise **clearly noted**. You may discuss solutions with your classmates but stop short of sharing your code with them. We check for plagiarism in a few ways and if your code is similar you will get a 0 on the assignment.

Academic Honesty

All work submitted in this course must be your own. Cheating and plagiarism will not be tolerated. If you have any questions about a specific case, *please ask me*.

NYU Poly's Policy on Academic Misconduct:

<http://engineering.nyu.edu/academics/code-of-conduct/academic-misconduct>

Course schedule (**Tentative**)

The course schedule is tentative, it's likely to change as the weeks go on.

In the following, OSTEP stands for **OSTEP** stands for **Operating Systems Three Easy Pieces**, MOS stands for "**Modern Operating Systems**", xv6 stands for "**xv6: A simple, Unix-like teaching operating system**"

Theme	Topic	Reading
Introductory Topics	Course Information. OS Intro.	OSTEP 1 xv6 0-1
	C Language Review	

	OS Overview	MOS 1 & xv6 0-1
	PC Hardware & Assembly Language	MOS 1 & xv6 0-1
	PC Hardware & Assembly Language	MOS 1 & xv6 0-1
	The Boot Process	xv6 Appendix B
	System Call Interface	xv6 Appendix B
	Processes	OSTEP 4, MOS 2 & xv6 5
	Process API	OSTEP 5, MOS 2 & xv6 5
	Scheduling Algorithms	OSTEP 7, MOS 2 & xv6 5
	Memory Management	MOS 3.3-3.5 & xv6 2
	Memory Management	MOS 3.3-3.5 & xv6 2
Memory Virtualization	Virtual Memory	MOS 3.3-3.5 & xv6 2
	Virtual Memory	MOS 3.3-3.5 & xv6 2
	Virtual Memory	MOS 3.3-3.5 & xv6 2
	Midterm	
Concurrency	Threads	MOS 3.1-3.2 & xv6 2
	Threads	MOS 2.3 – 2.5
	Concurrency	MOS 2.3 – 2.5. OSTEP 26-33
	Concurrency	
	Concurrency	

Persistence	Storage & Filesystems	MOS 4.1-4.4 & xv6 6
	Storage & Filesystems	MOS 4.1-4.4 & xv6 6
	Device Drivers, Interrupts, and I/O	MOS 5 & xv6 3
	Hard Disk Drives	OSTEP 37
Advanced Topics	RAID	OSTEP 38
	Security	MOS 9
	Virtualization	MOS 7
	Final Exam (Final date is decided schoolwide and assigned to each class, so please don't make travel plans before the end of finals week.	

Grading

Grading will be based on the following weights.

20% Homework

35% Midterm

35% Final Exam

10% Participation (polls during or after class)

Letter Grades

#	Symbol*	Start %*	Color	Assigned Value %
1	F	0		0
2	D	59.45		60
3	D+	66.45		67
4	C-	69.45		70
5	C	72.45		73
6	C+	76.45		77
7	B-	79.45		80
8	B	82.45		83
9	B+	86.45		87
10	A-	89.45		90
11	A	94.45		95

Other Grading notes:

Please take the following into consideration during and after the semester and save yourself one or many emails.

- 1) **I must grade every student EXACTLY the same way.** To this end, I cannot give you special consideration as a result of your academic status (probation or otherwise), scholarships, work status, family situation, visa status, race, color, creed, religious beliefs, past alien abductions, current moon cycle, location of the sun in the sky or anything other than your academic performance. **Your grade must be based on your academic performance in my class.**
- 2) **I cannot change your grade simply because you ask me to.** Your grade is calculated based on your performance from the first day of class to moment you turn in the final exam.
- 3) **I will not give you additional work.** Please remember that I must treat all students the same, so if I give you additional work, I would have to give it to the entire class. This is unfair to the students who complete their work on time.
- 4) **Your grade is a measure of your performance in my class.** If you receive an "F" it is because you have demonstrated that you do not understand the material in the course; if you receive an "A" it is because you have demonstrated that you

fully understand the material covered in the course. Other grades are assigned accordingly.

Moses Center Statement of Disability

If you are student with a disability who is requesting accommodations, please contact New York University's Moses Center for Students with Disabilities (CSD) at 212-998-4980 or mosescsd@nyu.edu. You must be registered with CSD to receive accommodations. Information about the Moses Center can be found at www.nyu.edu/csd. The Moses Center is located at 726 Broadway on the 3rd floor.

Academic Honesty

NYU School of Engineering Policies and Procedures on Academic Misconduct – complete Student Code of Conduct [here](#)

- A. **Introduction:** The School of Engineering encourages academic excellence in an environment that promotes honesty, integrity, and fairness, and students at the School of Engineering are expected to exhibit those qualities in their academic work. Through the process of submitting their own work and receiving honest feedback on that work, students may progress academically. Any act of academic dishonesty is seen as an attack upon the School and will not be tolerated. Furthermore, those who breach the School's rules on academic integrity will be sanctioned under this Policy. Students are responsible for familiarizing themselves with the School's Policy on Academic Misconduct.
- B. **Definition:** Academic dishonesty may include misrepresentation, deception, dishonesty, or any act of falsification committed by a student to influence a grade or other academic evaluation. Academic dishonesty also includes intentionally damaging the academic work of others or assisting other students in acts of dishonesty. Common examples of academically dishonest behavior include, but are not limited to, the following:
 - 1. **Cheating:** intentionally using or attempting to use unauthorized notes, books, electronic media, or electronic communications in an exam; talking with fellow students or looking at another person's work during an exam; submitting work prepared in advance for an in-class examination; having someone take an exam for you or taking an

exam for someone else; violating other rules governing the administration of examinations.

2. **Fabrication:** including but not limited to, falsifying experimental data and/or citations.
3. **Plagiarism:** Intentionally or knowingly representing the words or ideas of another as one's own in any academic exercise; failure to attribute direct quotations, paraphrases, or borrowed facts or information.
4. **Unauthorized collaboration:** working together on work meant to be done individually.
5. **Duplicating work:** presenting for grading the same work for more than one project or in more than one class, unless express and prior permission has been received from the course instructor(s) or research adviser involved.
6. **Forgery:** altering any academic document, including, but not limited to, academic records, admissions materials, or medical excuses.

NYU School of Engineering Policies and Procedures on Excused Absences – complete policy [here](#)

- A. Introduction: An absence can be excused if you have missed no more than **10 days of school**. If an illness or special circumstance has caused you to miss more than two weeks of school, please refer to the section labeled Medical Leave of Absence.
- B. Students may request special accommodations for an absence to be excused in the following cases:
 1. Medical reasons
 2. Death in immediate family
 3. Personal qualified emergencies (documentation must be provided)
 4. Religious Expression or Practice

Deanna Rayment, deanna.rayment@nyu.edu, is the *Coordinator of Student Advocacy, Compliance and Student Affairs* and handles excused absences. She is located in 5 MTC, LC240C and can assist you should it become necessary.

NYU School of Engineering Academic Calendar – complete list [here](#).

Please pay attention to notable dates such as Add/Drop, Withdrawal, etc. For confirmation of dates or further information, please contact Susana: sgarcia@nyu.edu